

Equivariant Algebra

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Previously On...

N_∞ & Twisted Products

- N_∞ operads are the same thing as indexing / transfer systems
- N_∞ operads parameterize (uniquely) “twisted” products

$$\mathrm{Map}(T, X) \rightarrow X$$

for admissible T .

Today's Goals

- 1 How does these structures interact?
- 2 What about in algebra / spectra?

Equivariant Algebra

Finite Coproduct Completion

$\mathcal{F}in^G$ is the finite coproduct completion of $\mathcal{O}rb^G$:

$$\mathrm{Fun}^{\sqcup}(\mathcal{F}in^G, \mathcal{C}) \xrightarrow{\cong} \mathrm{Fun}(\mathcal{O}rb^G, \mathcal{C}).$$

Rewriting Coefficient Systems

$$\mathcal{C}oeff \simeq \mathrm{Fun}^{\Pi}(\mathcal{F}in^{G,op}, \mathcal{S}et)$$

Coefficient Systems of Sets, Groups, Abelian Groups

If X is pointed,

$$\pi_k(X)(T) := \pi_k\left(\mathrm{Map}^G(T, X)\right).$$

Adjunctions

Adjunctions

We have an adjunction

$$G \times_H (-): \mathcal{F}in^H \rightleftarrows \mathcal{F}in^G: i_H^*.$$

Corollary

If $\mathcal{C}$ is a category with finite products, then we have an adjunction

$$i_H^*: \mathcal{C}oeff^G(\mathcal{C}) \rightleftarrows \mathcal{C}oeff^H(\mathcal{C}): \text{Colnd}_H^G.$$

Explicit Formulae

$$i_H^* \underline{M}(T) := \underline{M}(G \times_H T) \text{ \& \; } \text{Colnd}_H^G \underline{M}(T) := \underline{M}(i_H^* T).$$

Example: C_p -Coefficient Systems

$$\begin{array}{ccccc}
 0 & \longrightarrow & \underline{M}(C_p/C_p) & \xrightarrow{\text{Res}} & \underline{M}(C_p/e) \\
 \downarrow & & \downarrow \text{res} & & \downarrow \Delta \\
 \mathbb{Z}[G] \otimes \underline{M}(C_p/e) & \longrightarrow & \underline{M}(C_p/e) & \longrightarrow & \text{Hom}(\mathbb{Z}[G], \underline{M}(C_p/e)) \\
 \downarrow \gamma & & \downarrow \gamma & & \downarrow \gamma \\
 \text{Ind}_e^{C_p}(i_e^* \underline{M}) & & \underline{M} & & \text{CoInd}_e^{C_p}(i_e^* \underline{M})
 \end{array}$$

Contravariance

Definition

If T is a finite G -set and $\underline{M}$ is a coefficient system, let $\underline{M}_T$ be the coefficient system

$$S \mapsto \underline{M}(T \times S).$$

Proposition

This gives a diagram $\mathcal{F}in^{G,op} \rightarrow \mathcal{C}oeff^G$.

Extending Composites

Have a natural isomorphism of functors

$$\underline{M}_{G/H} \cong \text{CoInd}_H^G i_H^* \underline{M}.$$

Consequence of the Diagram

Evaluating at a point

Have $G/H \rightarrow *$ for any H :

- 1 Get a map

$$\underline{M}(*) \rightarrow \underline{M}_{G/H}(*) = \underline{M}(G/H)$$

- 2 and this is $\text{Aut}^G(G/H) \cong W_G(H)$ -equivariant, hence factors through fixed points

$$\underline{M}(*) \rightarrow \underline{M}(G/H)^{W_G(H)}.$$

Observation

Have representables h_T , and this is secretly the Yoneda lemma here.

$\mathcal{D}(V)$ -algebras

Transfers Give Transfers

If T is an admissible set for $\mathcal{O}$, then $\pi_k(X)$ has a T -transfer:

$$\pi_k(X)_T \rightarrow \pi_k(X).$$

Example

This gives us a map

$$\pi_k(X)(C_p/e) \rightarrow \pi_k(X)(C_p/C_p)$$

that

- 1 factors through the coinvariants
- 2 restricts to the trace.

Mackey Functors

Proposition

If X is a $\mathcal{D}(\infty\rho)$ -algebra in spaces, then the functor on $\mathcal{F}in^G, \cong$

$$T \mapsto \pi_k(X)_T$$

assembles to a covariant functor

$$\mathcal{F}in^G \rightarrow \mathcal{C}oeff^G.$$

Mackey Functors

This is the *external* form of a Mackey functor.

Lindner Category

Definition

Let $\mathcal{A}$ be the category with objects finite G -sets and with morphisms

$$\mathcal{A}(S, T) = \{S \leftarrow U \rightarrow T\} / \cong .$$

Composition is via pullback.

Definition

A Mackey functor is a product preserving functor

$$\underline{M}: \mathcal{A} \rightarrow \text{Set}.$$

Proposition

$\underline{M}(T)$ is always a commutative monoid & all structure maps are monoid maps.

Example

A C_p -Mackey functors is a diagram:

$$\begin{array}{ccc}
 & \underline{M}(C_p/C_p) & \\
 \text{res} \swarrow & & \searrow \text{tr} \\
 & \underline{M}(C_p/e) & \\
 & \uparrow \gamma &
 \end{array}$$

satisfying $\text{res} \circ \text{tr}(m) = \sum_{g \in C_p} g \cdot m$, $\gamma \circ \text{res} = \text{res}$, $\text{tr} \circ \gamma = \text{tr}$.

Example

$$\underline{A}_T := \mathbb{Z} \bigotimes_{\mathbb{N}} \mathcal{A}(T, -).$$

Incomplete Mackey Functors

Definition

An indexing category for G is a subcategory $\mathcal{F}in_O^G$ of $\mathcal{F}in^G$ that is

- 1 wide
- 2 finite coproduct complete
- 3 pullback stable.

Proposition

The poset of indexing categories on G is $\text{Tran}(G)$.

Incomplete Mackey Functors

Definition

The $\mathcal{O}$ -Lindner category has objects finite G -sets and

$$\mathcal{A}^{\mathcal{O}}(S, T) = \{S \xleftarrow{f} U \xrightarrow{g} T \mid g \in \mathcal{F}in_{\mathcal{O}}^G\} / \cong .$$

Definition

An $\mathcal{O}$ -Mackey functor is a product preserving functor $\mathcal{A}^{\mathcal{O}} \rightarrow \mathcal{A}b$.

Theorem

If $\mathcal{O}$ is an N_{∞} -operad, then for all k , $\pi_k(X)$ is naturally an $\mathcal{O}$ -Mackey functor.

Multiplicative Version

Observation

We have a distributive isomorphism

$$\bigotimes_{i \in I} \bigoplus_{j \in J} X_{i,j} \cong \bigoplus_{f \in \text{Map}(I,J)} X^{\otimes f},$$

where

$$X^{\otimes f} = \bigotimes_{i \in I} X_{i,f(i)}.$$

Main Slogan!

Equivariantize all these things!

Dependent Product

LCC

Let $f: S \rightarrow T$ in $\mathcal{F}in^G$. The pullback

$$f^*: \mathcal{F}in^G_T \rightarrow \mathcal{F}in^G_S$$

has both adjoints:

- L Dependent sum Σ_f
- R Dependent product Π_f .

Corollary

Have a natural transformation of endofunctors of $\mathcal{F}in^G_S$:

$$\epsilon_f: f^* \circ \Pi_f \Rightarrow Id.$$

Exponential Diagrams

Definition

An exponential diagram is one isomorphic to

$$\begin{array}{ccccc}
 S & \xleftarrow{h} & S' & \xleftarrow{f' = \epsilon_g(h)} & S \times_T \Pi_f(S') \\
 g \downarrow & & & & \downarrow g' = h'^*(g) \\
 T & \xleftarrow{h' = \Pi_f(h)} & & & \Pi_f(S').
 \end{array}$$

Slogan III

Exponential diagrams encode distributivity

Polynomials

Definition

Let $\mathcal{P}$ be the category with objects finite G -sets and with morphisms “polynomials”

$$T_h N_g R_f := [S \xleftarrow{f} U \xrightarrow{g} V \xrightarrow{h} T]$$

Composition Rules

① $R_f \circ R_{f'} = R_{f' \circ f}$, $T_h \circ T_{h'} = T_{h \circ h'}$, and $N_g \circ N_{g'} = N_{g \circ g'}$.

② If we have a pullback diagram

$$\begin{array}{ccc} S' & \xrightarrow{g'} & T' \\ f' \downarrow & & \downarrow f \\ S & \xrightarrow{g} & T \end{array}$$

$$R_f \circ T_g = T_{g'} \circ R_{f'} \text{ and } R_f \circ N_g = N_{g'} \circ R_{f'}.$$

Polynomials (Continued)

Composition Rules Continued

③ If we have an exponential diagram

$$\begin{array}{ccccc}
 S & \xleftarrow{h} & S' & \xleftarrow{f' = \epsilon_g(h)} & S \times_T \Pi_f(S') \\
 g \downarrow & & & & \downarrow g' = h'^*(g) \\
 T & \xleftarrow{h' = \Pi_f(h)} & & & \Pi_f(S').
 \end{array}$$

then we have

$$N_g \circ T_h = T_{h'} \circ N_{g'} \circ R_{f'}.$$

TNR Functors

Proposition

The disjoint union of finite G-sets is the categorical product in $\mathcal{P}$.

Definition

A TNR / Tambara functor is a product preserving functor $\underline{R}: \mathcal{P} \rightarrow \text{Set}$.

Unpacking

Have semirings $\underline{R}(G/H)$ for all H plus, for any $G/H \rightarrow G/K$,

- ① A semi-ring map $\text{res}_H^K: \underline{R}(G/K) \rightarrow \underline{R}(G/H)$,
- ② a map of $\underline{R}(G/K)$ -modules $\text{tr}_H^K: \underline{R}(G/H) \rightarrow \underline{R}(G/K)$, and
- ③ a multiplicative map $n_H^K: \underline{R}(G/H) \rightarrow \underline{R}(G/K)$.

How the ...

Practical Reduction I

Disjoint Union and induction are disjunctive: disjoint union gives an isomorphism

$$\text{Map}^G(G/H, T) \amalg \text{Map}^G(G/H, T') \cong \text{Map}^G(G/H, T \amalg T'),$$

and any map $f: T \rightarrow G/H$ is isomorphic to

$$G \times_H f^{-1}(eH) \rightarrow G \times_H * \cong G/H.$$

Corollary

*This reduces our exponential diagrams to consider to $T = *$.*

Further Reductions

Practical Reduction II

- 1 If $g = * \amalg * \rightarrow *$ is the fold, and $h: S \amalg T \rightarrow * \amalg *$, then

$$\Pi_g(h) = S \times T \rightarrow *.$$

- 2 If $g = G/H \rightarrow *$, and $h: G \times_H S \rightarrow G/H$, then

$$\Pi_g(h) = \text{Map}^H(G, S).$$

Corollary

Have formulas like

$$n_e^{C_2}(a + b) = n_e^{C_2}(a) + n_e^{C_2}(b) + tr_e^{C_2}(a\bar{b}).$$

bi-Incomplete Tambara Functors

Definition

If $\mathcal{O}_a$ and $\mathcal{O}_m$ are two transfer systems, let $\mathcal{P}^{(\mathcal{O}_a, \mathcal{O}_m)}$ be the wide subgraph of $\mathcal{P}$ with

$$\mathcal{P}^{(\mathcal{O}_a, \mathcal{O}_m)}(S, T) = \{S \xleftarrow{f} U \xrightarrow{g} V \xrightarrow{h} T \mid g \in \mathcal{F}in_{\mathcal{O}_m}^G, h \in \mathcal{F}in_{\mathcal{O}_a}^G\} / \cong .$$

Definition

We say a pair $(\mathcal{O}_a, \mathcal{O}_m)$ is compatible if $\mathcal{P}^{(\mathcal{O}_a, \mathcal{O}_m)}$ is a subcategory.

Example

$(\mathcal{O}^{gen}, \mathcal{O})$ is compatible for any $\mathcal{O}$.

Compatibility

Theorem (Blumberg–H.)

$(\mathcal{O}_a, \mathcal{O}_m)$ is compatible iff for all $H/K \in \underline{\mathcal{O}}_m(H)$, we have that for all $T \in \underline{\mathcal{O}}_a(K)$,

$$\text{Map}^K(H, T) \in \underline{\mathcal{O}}_a(H).$$

Theorem (Chan)

$(\mathcal{O}_a, \mathcal{O}_m)$ is compatible if we can complete any solid diagram

$$\begin{array}{ccc} H & \xrightarrow{\mathcal{O}_m} & K \\ \uparrow \mathcal{O}_a & & \uparrow \mathcal{O}_a \\ H \cap J & \xrightarrow{\mathcal{O}_m} & J \end{array}$$

Externalizing Tambara Functors

Internal vs External Mackey Functors

Had an equivalence between Mackey functors and extensions

$$\begin{array}{ccc}
 \underline{\mathcal{F}in}^{\cong} & \xrightarrow{T \mapsto \underline{M}_T} & \underline{Coeff.} \\
 \downarrow & \nearrow \text{dotted} & \\
 \underline{\mathcal{F}in} & &
 \end{array}$$

Question

- 1 *How can we simplify this?*
- 2 *How do we fold in the twisted tensor product?*

G-Commutative Monoid

Restriction

We have a natural isomorphism

$$i_H^*(\underline{M}_T) \cong i_H^* \underline{M}_{i_H^* T}$$

Transfers Restrict

If T is an admissible G -set for $\mathcal{D}(V)$, then the restriction of

$$\pi_k(X)_T \rightarrow \pi_k(X)$$

is

$$i_H^*(\pi_k(X))_{i_H^* T} \rightarrow i_H^*(\pi_k(X)).$$

Aside: Overdetermination

Remark

If T is an H -set, then the map $(i_H^ \underline{M})_T \rightarrow i_H^* \underline{M}$ is a retract of the restriction to H of*

$$\underline{M}_{G \times_H T} \rightarrow \underline{M}_{G/H}.$$

Corollary

A Mackey functor is equivalently an extension

$$\begin{array}{ccc} \mathcal{F}in^{G, \cong} & \xrightarrow{T \mapsto \underline{M}_T} & \text{Coeff}^G \\ \downarrow & \nearrow & \\ \mathcal{F}in^G & & \end{array}$$